

Sudhanva Paturkar

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Summary

Full-stack engineer at Adobe working across React and TypeScript frontends and Java/Spring Boot services: document editing surfaces, generative-AI product features, personalization pipelines, and the testing and observability around them. Supplements that with hands-on agentic AI work: LLM agent loops with guardrails, structured-output validation, provider abstraction, and evaluation harnesses.

Experience

Adobe

May 2025 - Present | San Jose, CA

Software Engineer II

- Built the frontend surfaces for **AI Voice Over**, a generative-AI feature in Adobe Acrobat, and worked with the routing and agent teams on its agent integration. Handled asynchronous, long-running agent execution on the client.
- Built the quizzes, flashcards, and study packet components behind **Study Spaces**, implemented within the Jot editor library used to render note documents in Acrobat.
- Developed an end-to-end extensible toolbar for the Jot editor, including the APIs it exposes, used across mobile, desktop, and web surfaces.
- Led frontend development for feature discovery in Acrobat Web using React and TypeScript, and instrumented it with the Adobe Analytics SDK and custom event tracking to measure its rollout.

Adobe

June 2024 - August 2024 | San Jose, CA

Software Engineer Intern

- Built a personalized, user-data-driven notification system for Adobe Acrobat in Java and Spring Boot that analyzed behavior patterns and increased targeted feature adoption by 15% through customized in-app messaging.
- Contributed to a Spring Boot microservice processing over 5 million daily log entries with batch processing and multi-threading to generate the user behavior profiles powering the personalization engine.
- Worked with Apache Kafka pipelines and wrote custom producers and consumers to stream data in real time between distributed services.

Accenture

June 2021 - May 2023 | Pune, India

Senior Full-Stack Developer

Promoted from Full-Stack Developer, December 2022

- Built document-processing API functionality covering PDF manipulation, digital signature verification, and document versioning, along with Splunk dashboards to monitor API performance and system health metrics.
- Integrated in-house AI document-processing APIs into the React frontend and rendered their analysis and extraction results in the product UI.
- Designed and implemented automated test suites with JUnit4 for backend Java services and Selenium WebDriver for React components, achieving 85% code coverage and a 20% decrease in production regressions.
- Resolved critical customer-reported defects in an enterprise document-signing platform (Java Spring Boot, React) and drove performance work spanning query optimization, caching, and async API polling. Awarded the Accenture Pinnacle Award for contributions to Product Engineering, FY22-Q4.

Projects

CatanCards: Agentic Harness for LLM-Driven Players

August 2026

- Built an agentic harness that drives autonomous LLM players in a live multiplayer Node.js application, with a decision loop built on single-flight concurrency control, progress-based termination guards, and per-turn action budgets.
- Modeled the harness as an explicit state machine that routes each decision point to a specialized policy, with conditional edges, guarded transitions, and per-run persistence implemented directly on Node.js.
- Enforced structured outputs by constraining the model to a runtime-generated menu of state-legal actions, then added payload validation and a deterministic policy fallback that keeps the agent available on malformed or off-menu responses.
- Designed a provider-agnostic LLM client over Ollama and Google Gemini with per-call-type timeout budgets, typed failure reasons, and a health endpoint tracking call volume, latency, and failure rate; sustained 487 calls at a 2.8s mean with a single fallback across a 128-turn run.
- Built an offline evaluation harness over per-decision trajectory logs that scores optimal-choice rate, model-vs-fallback split, and repeated-justification detection to flag agent regressions between runs.

Study Buddy: AI Learning Assistant [Hackathon Winner]

April 2025

- Engineered an AI study platform in React, Node.js, and Express that used the Gemini 2.5 Pro API to turn unstructured PDFs and video transcripts into structured outputs: interactive mind maps, study schedules, and hierarchical topic breakdowns.
- Designed chain-of-thought prompt pipelines that returned schema-constrained responses, so the frontend rendered model output directly instead of parsing free text.
- Orchestrated a specialized agent per output type as a LangGraph fan-out and gather graph, ran the branches concurrently, and reconciled their asynchronous completions so each section of a study packet rendered independently.
- Implemented an output evaluation pass validating schema conformance and source coverage, with automatic regeneration on failed checks.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, SQL

AI & Agentic Systems: LLM model APIs (Google Gemini, Ollama / local models), tool use & structured outputs, prompt engineering, LangGraph, agent loops & tool routing, guardrails, evaluation harnesses & trajectory logging

Frameworks: Node.js, React, Express.js, Spring Boot, Socket.IO

Platform & Cloud: AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD pipelines, Apache Kafka

Databases: PostgreSQL, MySQL

Testing & Observability: Jest, Cypress, JUnit4, Selenium WebDriver, React Testing Library, Splunk

Certifications: AWS Certified Developer - Associate

Education

Khoury College of Computer Sciences, Northeastern University

September 2023 - April 2025

Master of Science in Computer Science, GPA - 3.91

Nagpur University

August 2017 - June 2021

Bachelor of Engineering in Computer Technology, GPA - 3.71